TaxonVision

Production Biodiversity MLOps Architecture

High-Throughput Vision · Split Conformal Safety · Automated Closed Loop



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Final MLOps Project Presentation & Architecture Defense

The Mission: Biodiversity MLOps

Why Real-World Data Demands Resilient Systems

The Ecological Problem

- **Biodiversity Crisis:** Tracking species loss requires classifying millions of wild citizen-science observations at global scale.
- **Why Standard ML Fails in Nature:**
  - Extreme long-tail class imbalance (80% rare taxa).
  - Continuous visual drift (lighting, camera sensors, seasons).
  - Severe risk of silent false positives on unknown taxa.

The High Cost of Hallucinations:

Misclassifying an endangered organism as common ruins survey datasets and compromises conservation policy.

Our Core Engineering Mission

- **1. Strict Data Reproducibility:** End-to-end versioned DVC DAG from raw S3 open data to registered model checkpoints.
- **2. Epistemic Safety Envelopes:** Mathematical coverage guarantee ($1 - \alpha \geq 95\%$) via Split Conformal Prediction + OOD gating.
- **3. Self-Healing Retraining Loop:** Continuous feature drift detection (Evidently AI) triggering automated retraining pipelines.

Production Objective: A hardened, self-monitoring MLOps engine operating safely in the wild.

Curriculum & System Roadmap

Three Core Phases & Architectural Milestones

Phase 1: Foundation

Data Plane & Reproducibility

- **S3 Streaming:** Direct ingest from open data bucket.
- **Licence Audit:** Creative Commons filter (CC0, CC-BY, CC-BY-NC).
- **DVC 7-Stage DAG:** Strict MD5 tracking & zero split leakage.
- **Feature Caching:** Pre-computed DINOv2 embeddings on NVMe.
- **Bayesian HPO:** Optuna sweeps with MLflow tracking.

Phase 2: Safety & Serving

Low-Latency Inference

- **INT8 Quantization:** Dual-output ONNX (< 15 ms CPU SLO).
- **Conformal Sets:** Finite-sample quantile sets ($1 - \alpha \geq 95\%$).
- **Helmholtz Energy:** Logit-based out-of-distribution rejection.
- **Dual Serving:** REST API and Jinja2/HTMX dashboard.
- **Explainability:** On-demand Forward-Only CAM heatmaps.

Phase 3: Closed Loop

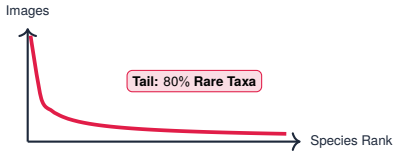
Drift, Alerts & Retraining

- **Vector Telemetry:** Async PostgreSQL logging of 384-D vectors.
- **Real-Time Cockpit:** Prometheus TSDB + 6-Panel Grafana.
- **Drift Detection:** Evidently AI W_1 & Kernel MMD^2 tests.
- **Auto Retraining:** Grafana Webhook Relay $\rightarrow$ Airflow 3.0.
- **Active Learning:** Label Studio triage for uncertain sets.

The Domain Challenge

Why Standard Computer Vision Fails in the Wild

Extreme Long-Tail Distribution



High Cost of Hallucinations

Misclassifying an endangered organism as a common species invalidates biodiversity surveys and damages conservation.

Three Production Realities

- **Severe Class Imbalance:** A few common taxa have 10,000+ photos; thousands of rare species have < 10.
- **Continuous Feature Drift:** Lighting, seasonal changes, camera sensors, and geographic shifts constantly distort the visual manifold.
- **Novelty & Noise:** Users submit unseen species, blur, and blank shots – the model must abstain!

The MLOps Paradigm Shift

From Notebook Scripts to Production Systems

Prototype / Toy Machine Learning

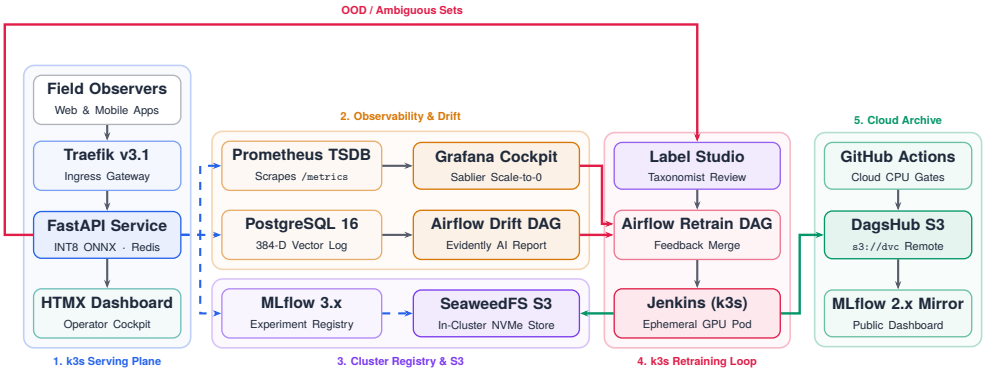
- **Unversioned Data:** Ad-hoc CSVs, non-reproducible manual train/test splits.
- **Redundant Compute:** Heavy vision backbones re-extracted on every run.
- **Raw Softmax Guesses:** Uncalibrated probabilities, forced to predict on unknown classes.
- **Silent Decay:** Zero drift monitoring; model silently fails in production.

TaxonVision Production MLOps

- **DVC Versioned DAG:** 7 pure stages cryptographically tracked by MD5 hashes.
- **Zero-Copy Feature Caching:** Pre-computed foundation embeddings cached on NVMe.
- **Conformal Safety Sets:** Guaranteed coverage ($1 - \alpha \geq 95\%$) with Energy OOD filter.
- **Closed Retraining Loop:** Automated alerts $\rightarrow$ Airflow DAG $\rightarrow$ GPU retraining.

Complete MLOps Infrastructure

Unified Serving, Telemetry, Storage & Retraining



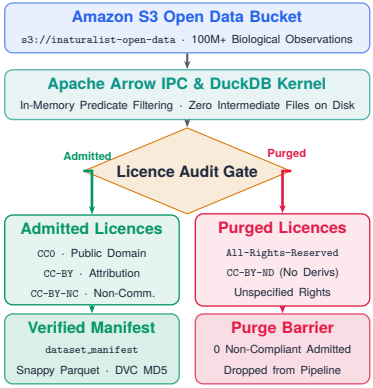
Data Plane & Licence Governance

Zero-Copy S3 Streaming & Open Licences

Zero-Copy Ingestion Flow

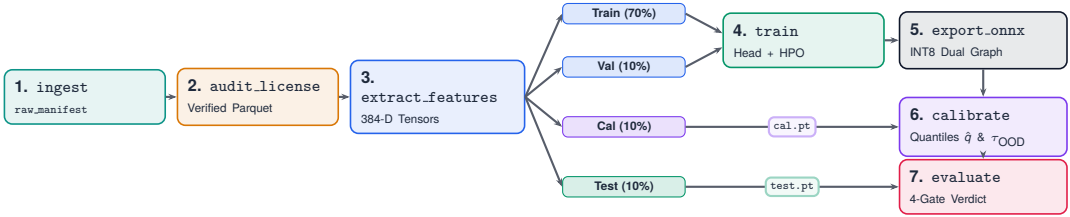
- **Direct S3 Streaming:** Ingests batches from `s3://inaturalist-open-data` without downloading full tarballs.
- **In-Memory Arrow IPC:** DuckDB filters observations in RAM; image buffers are processed ephemerally.
- **Attribution Invariant:** Retains photographer name and licence code across every manifest stage.

Governance Guarantee: Only verified Creative Commons assets enter the Epistemic DAG. Zero unverified records admitted.



The Epistemic Diamond DAG

7-Stage DAG with Zero Split Leakage



Epistemic Diamond Split Invariant: **Train** updates head weights; **Val** guides Optuna early stopping; **Cal** quantiles conformal coverage; **Test** renders final promotion verdict. Zero leakage!

Dual-Tier Storage Topology

DagsHub Cloud S3 vs. Local NVMe Store

Tier 1: Cloud Collaboration (DagsHub)

- **Remote:** Hosted DagsHub S3 (`s3://dvc`).
- **Role:** Immutable public archive for manifests, embeddings, and registered ONNX models.
- **CI Integration:** GitHub Actions pulls checkpoints using `DAGSHUB_TOKEN` without cluster access.
- **Open Science:** Public MLflow experiment tracker.

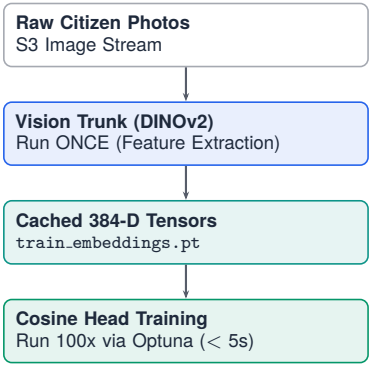
Tier 2: In-Cluster Storage (Workstation)

- **Remote:** Local SeaweedFS S3 (`local-seaweedfs`).
- **Role:** High-throughput NVMe-backed storage for embeddings and local model staging.
- **Zero Egress Costs:** 100,000s of feature embeddings synced locally without cloud bandwidth bills.
- **Private MLflow 3.x:** High-frequency Optuna logging.

Feature Caching & Rapid Training

Decoupled Vision Architecture

- **Frozen Trunk:** Foundation vision backbone (DINOv2 ViT-Small, $D = 384$).
- **Lightweight Head:** Cosine-similarity classification head mapping embeddings to species logits.
- **The Feature Caching Strategy:**
  - Forward-passing through the 22M parameter ViT takes minutes.
  - Once pre-computed, visual representations never change!
  - Extract features **once**; cache `.pt` tensors on NVMe.
- **Result:** Head training runs in < 5 **seconds**!

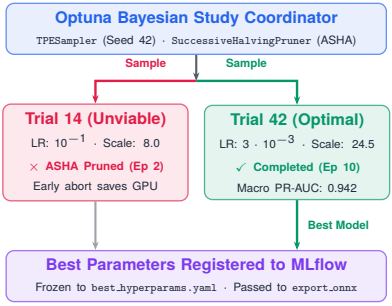


Bayesian Tuning: Optuna

TPE Search & ASHA Pruning on Cached Manifolds

Bayesian Hyperparameter Search

- **TPE Sampler (Parzen Estimator):** Models $P(x \mid y)$ to steer proposals toward high PR-AUC regions instead of random search.
- **ASHA Pruner (Successive Halving):** Terminates unviable trials at epoch 2, eliminating wasted compute on poor configurations.
- **Cached Embedding Synergy:** Training on cached 384-D tensors takes < 0.5 s/epoch; a 100-trial study finishes in < 3 **minutes**!
- **MLflow Registry Integration:** Every trial tracks params, PR-AUC, and pruning state; the optimal head is tagged for ONNX export.



Zero-Allocation INT8 Inference

Sub-15ms Low-Latency Inference Under SLOs

11.8 ms
CPU p95 (Budget: $\leq$ 15 ms)

6.2 ms
GPU p95 (Budget: $\leq$ 10 ms)

18.4 ms
API p95 (Budget: $\leq$ 25 ms)

Dual-Output ONNX Architecture

- Export outputs both classification logits and normalized 384-D feature embeddings in a **single forward pass**.
- Eliminates redundant forward passes for telemetry logging and drift analysis.

Zero-Allocation Discipline

- Pre-allocated contiguous C-order NumPy arrays for input and prediction buffers.
- Zero Python dictionary boxing or dynamic list allocations inside high-frequency serving loops.

Epistemic Safety: Softmax Failure

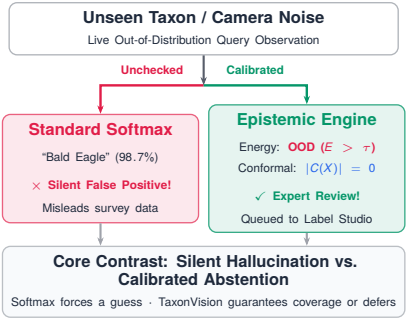
Why Deep Neural Networks Fail Silently in Nature

The Illusion of Softmax "Confidence"

- Softmax mathematically forces output probabilities to sum to 1.0 (100%).
- When presented with unseen organisms or camera glare, standard models **cannot abstain**.
- **Result:** High-confidence ($> 98\%$) silent hallucinations on completely wrong taxa!

The Operational Safety Requirement

- Production vision systems must reject false certainty and defer ambiguous cases to human experts.



Split Conformal Prediction Sets

Guaranteed Coverage Sets Instead of Guesses

The Conformal Prediction Guarantee

- Instead of forcing a single guessed species, the model outputs an **adaptive prediction set** $C(X)$.
- The user specifies a safety confidence: e.g. **95**% coverage ($\alpha = 0.05$).
- **Mathematical Guarantee:** The true species is mathematically guaranteed to be in the set $\geq$ 95% of the time:

$$P(Y \in C(X)) \geq 1 - \alpha$$

- Calibrated strictly on the holdout **Calibration split**.

Three Operational Set Outcomes

Case 1: Decisive ($|C| = 1$)
- Set: {`"Barn Owl"`}
- High confidence; automated response.

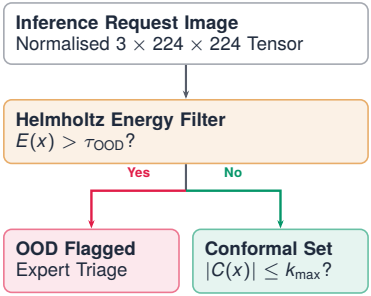
Case 2: Ambiguous ($1 < |C| \leq 3$)
- Set: {`"Common Gull"`, `"Herring Gull"`}
- Narrows search to sister species.

Case 3: Referral ($|C| > 5$ or $|C| = 0$)
- Model defers to human triage queue!

Out-of-Distribution Gating

Energy-Based OOD Gating

- Computes negative Helmholtz free energy $E(x)$ over unconstrained model logits.
- In-distribution taxa produce low energy; novel organisms, blurred photos, or blank frames produce high energy.
- Threshold τ_{OOD} calibrated to maintain 95% true positive rate on calibration data.
- Energy $>$ τ_{OOD} $\implies$ Immediate bypass of classification directly to human triage.

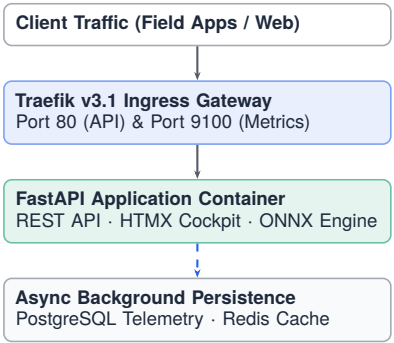


Unified Serving: REST API & HTMX

Dual-Interface Service Without Build Ballast

Unified Dual-Interface Architecture

- **Machine REST API:** Strongly typed Pydantic endpoints (`/api/v1/predict`, `/explain`, `/drift/evaluate`).
- **Operator Dashboard:** Server-rendered Jinja2 + HTMX cockpit at `/dashboard`; zero JavaScript build complexity.
- **Forward-Only CAM:** Generates gradient-free spatial projection heatmaps ($<$ 3 ms) on demand off the event loop thread.
- **Traefik Edge Reverse Proxy:** Handles SSL termination, load balancing, and rate limiting.

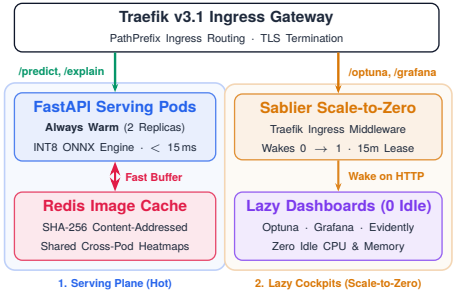


k3s Cluster Architecture

Scale-to-Zero via Sablier & Redis Image Cache

Lightweight Edge Orchestration

- **k3s Edge Cluster:**
  $<$ 512 MB control plane, Traefik v3.1 ingress controller, non-root container isolation.
- **Always-Warm Serving Plane:**
  FastAPI replicas keep INT8 ONNX engine pre-warmed for deterministic $<$ 15 ms inference.
- **Scale-to-Zero via Sablier:**
  Dashboards (Optuna, Grafana, Evidently) consume **0 CPU/RAM** when idle. Auto-wakes $0 \rightarrow 1$ on HTTP ingress with 15m session lease.
- **Distributed Redis Image Cache:**
  Shared SHA-256 buffer (`taxon:img:*`, LRU eviction) eliminates duplicate image transfers for Forward CAM heatmaps across pods.



Vector Telemetry & Persistence

Why Persist 384-D Embeddings?

- Detecting representation shift requires analysing feature manifolds from live traffic.
- Storing raw citizen photos wastes storage and violates privacy.
- TaxonVision persists the **384-dimensional feature vector** (`embedding REAL[]`) directly in PostgreSQL.
- **Event Loop Protection:** Written asynchronously via `BackgroundTasks` - request returns in 18 ms while I/O runs in background.

Telemetry Schema (`predictions` table)

- `observation_id` (UUID primary key)
- `created_at` (Timestamptz index)
- `top1_taxon` & `top1_confidence`
- `conformal_set` (`TEXT[]` array)
- `ood_energy` & `is_ood` (Boolean)
- `latency_ms` (Measured wall-clock)
- `embedding` (`REAL[]`, **384 dimensions**)

Ready for instantaneous sliding-window drift queries!

Automated Drift Detection

Two-Tier Hypothesis Testing & Evidently AI

Two-Tier Drift Evaluation Engine

- **Tier 1: Wasserstein-1 (W_1) on Norms**
  Fast $O(N)$ linear-time check on embedding lengths; catches immediate camera/contrast shift.
- **Tier 2: Kernel MMD² Permutation Test**
  Multivariate distribution test ($B = 200$ permutations); detects subtle manifold distortion ($p < 0.01$).
- **Evidently AI 0.7 HTML Dashboard:**
  Projects 384 dims to 8 PCA components and renders interactive reports (`drift_report.html`).

Automated MLflow Tracking Lineage

- Evaluated hourly by the Airflow DAG `prediction_drift_monitor`.
- Automatically creates an MLflow run in experiment `taxon-vision-drift`.
- Logs metrics (`drift_w1`, `drift_mmd2`, `p_value`), and tags `drift_detected=true/false`.
- Attaches the Evidently HTML report as an immutable build artifact.

Prometheus & Grafana Cockpit

Real-Time Operational Monitoring & Alerts

Real-Time Metrics & Dashboards

- **Prometheus Pull Scraper:** Scrapes `/metrics` every 15 seconds across serving and Traefik pods.
- **Grafana Cockpit (6 Panels):**
  - Request rate and status codes (2xx, 4xx, 5xx).
  - p50, p95, p99 end-to-end latency histograms.
  - Conformal set size distribution.
  - Human triage referral rate.
  - Live drift distance gauge (W_1).

Four Production Alerting Rules

- `PredictionLatencyP95High`: p95 latency $>$ 120 ms for 5 min (`warning`).
- `PredictionErrorRatioHigh`: Error ratio $>$ 5% for 5 min (`critical`).
- `EmbeddingDriftDetected`: Drift gauge $>$ 0.5 (`critical`, **action: retrain**).
- `DriftMonitorStale`: No drift run in $>$ 2 hours (`warning`).

Closed-Loop Retraining Trigger

Grafana Alert Webhook to Airflow 3.0 DAG



Anti-Alert-Storm Cooldown Protection: When sudden environmental drift causes continuous alerting, the atomic 6-hour cooldown window ensures the GPU cluster executes only one retraining run, preventing resource exhaustion.

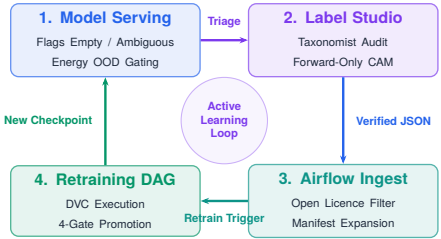
Human-in-the-Loop Active Learning

Active Learning & Review via Label Studio

The Curation Lifecycle

- **1. Epistemic Triage:**
  Ambiguous ($|C| > 3$) or OOD energy observations route to review queue.
- **2. Expert Verification:**
  Taxonomists inspect morphology & forward-only CAM in Label Studio.
- **3. Feedback Merge:**
  Airflow verifies open licences and expands training manifest.

Core Benefit: Focuses expensive human expert hours exclusively on borderline cases where the model lacks coverage.



Four-Gate Model Promotion Verdict

Automated Multi-Objective Quality Verification

Why Validation Accuracy Is Not Enough

- In production, a model can achieve high accuracy but fail latency SLOs, violate conformal coverage, or degrade under INT8 quantization.
- TaxonVision enforces a strict **Four-Gate Promotion Verdict** in Stage 7 (`evaluate.py`) before any model receives the `@Production` tag.

The Four Automated Promotion Gates

Gate 1: Predictive Superiority

- Validation Macro PR-AUC $\geq$ active production baseline.

Gate 2: Quantization Fidelity

- INT8 accuracy drop $\leq$ 3.0% relative to FP32 baseline.

Gate 3: Empirical Conformal Coverage

- Holdout test coverage $\geq$ 90.0% (nominal 95%).

Gate 4: Latency Ceiling & Exchangeability

- CPU p95 latency $\leq$ 120 ms and live MMD $p \geq 0.01$.

Summary: Three Pillars of MLOps

Foundations, Epistemic Safety & Automation

1. Robust MLOps

- k3s & Sablier Scale-to-Zero
- 7-Stage DVC Pipeline
- Dual-Tier S3 Remote
- MLflow 3.x Registry
- Pure Pixi Toolchain

2. Speed & Safety

- Sub-15ms INT8 ONNX
- Optuna TPE & ASHA
- Redis Distributed Cache
- 95% Coverage Guarantee
- Forward-Only CAM

3. Closed Loop

- Postgres 384-D Ingestion
- Evidently 0.7 Drift DAG
- Grafana Webhook Relay
- Label Studio Active Learn
- Jenkins k3s Ephemeral GPU

Operational Reliability Axiom: Real-world vision robustness is governed not by peak validation accuracy, but by deterministic epistemic safety envelopes: split conformal coverage guarantees, representation manifold drift tracking, and automated retraining pipelines.

Live Demonstration & Q&A

Real-Time Inference · Conformal Sets · Observability Cockpit

1. HTMX UI & CAM
2. Grafana Cockpit
3. MLflow Registry
4. Airflow Retraining

Thank you for your attention!

We now transition to the live system demonstration and jury questions.